TEAM NO:3

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PROJECT REPORT

Simulated Temperature Monitoring using ESP32 and Raspberry Pi (MQTT)

Abstract

This project demonstrates IoT based temperature monitoring using ESP32 and Raspberry Pi with MQTT protocol. The ESP32 acts as a publisher and sends simulated temperature data to the Raspberry Pi, which acts as broker and subscriber. The data is displayed on Raspberry Pi terminal using Python.

Objective

To establish MQTT communication between ESP32 and Raspberry Pi and monitor temperature wirelessly.

Components Required:

Hardware Required

- ✓ ESP32 Development Board
- ✓ Raspberry Pi
- ✓ USB Cable
- ✓ Laptop
- ✓ WiFi Connection

Software Required

- ✓ Arduino IDE
- ✓ Mosquitto MQTT Broker
- ✓ Python 3
- ✓ Paho MQTT Library

Procedure

1. Installed Mosquitto broker on Raspberry Pi.
2. Started Mosquitto service.
3. Found Raspberry Pi IP address.
4. Installed MQTT library in Arduino IDE.
5. Uploaded ESP32 publisher code.
6. Installed Python MQTT subscriber.
7. Executed subscriber code.
8. Verified temperature data received.

What is MQTT Broker?

An MQTT broker is a central server that receives messages from publishing clients and distributes them to subscribed clients based on topics.

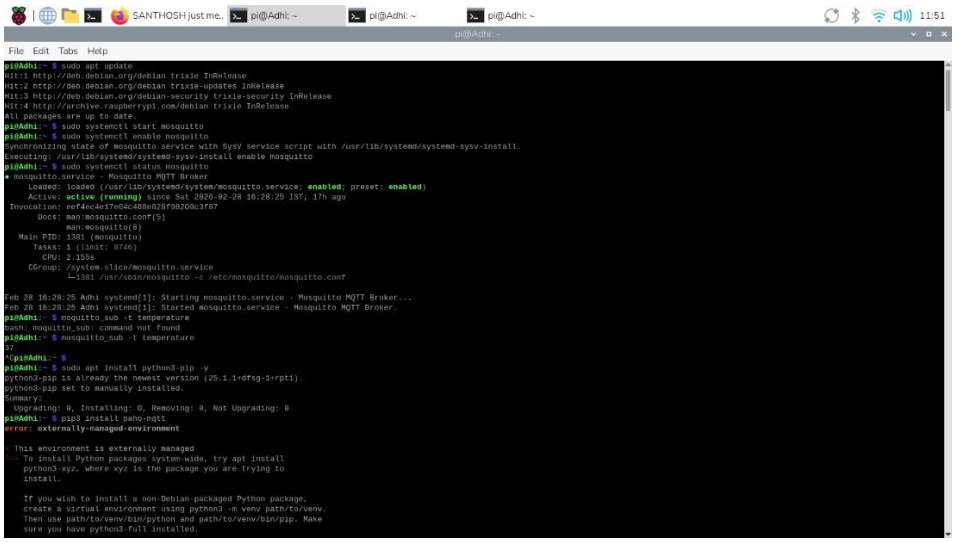
How it Works?

- ✓ A device (publisher) sends a message to a specific topic.
- ✓ The MQTT broker receives the message.
- ✓ The broker forwards the message to all devices (subscribers) that are subscribed to that topic.

Output

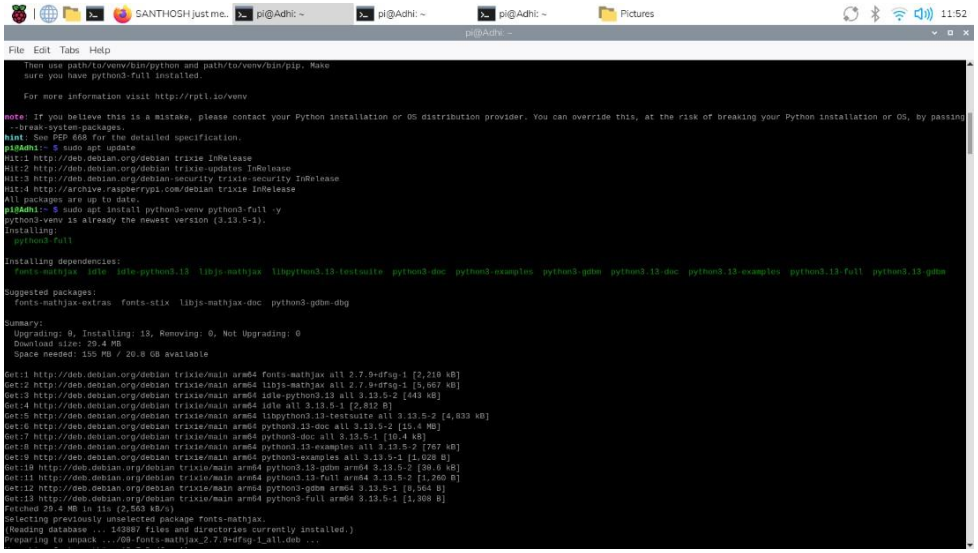
Step 1:

Installed and verified the Mosquitto MQTT broker on Raspberry Pi.



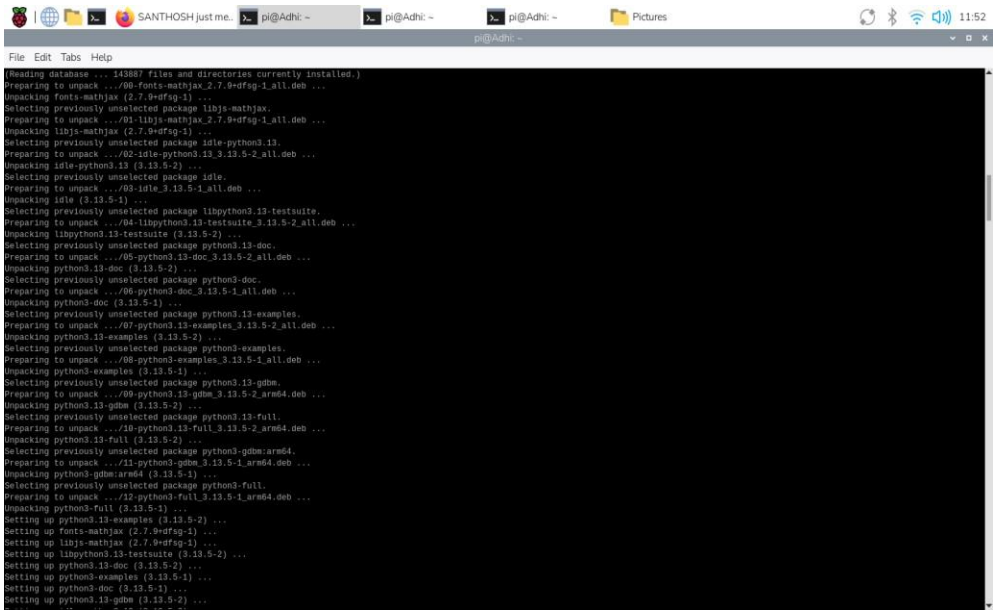
Step 2:

Started the Mosquitto broker service successfully.



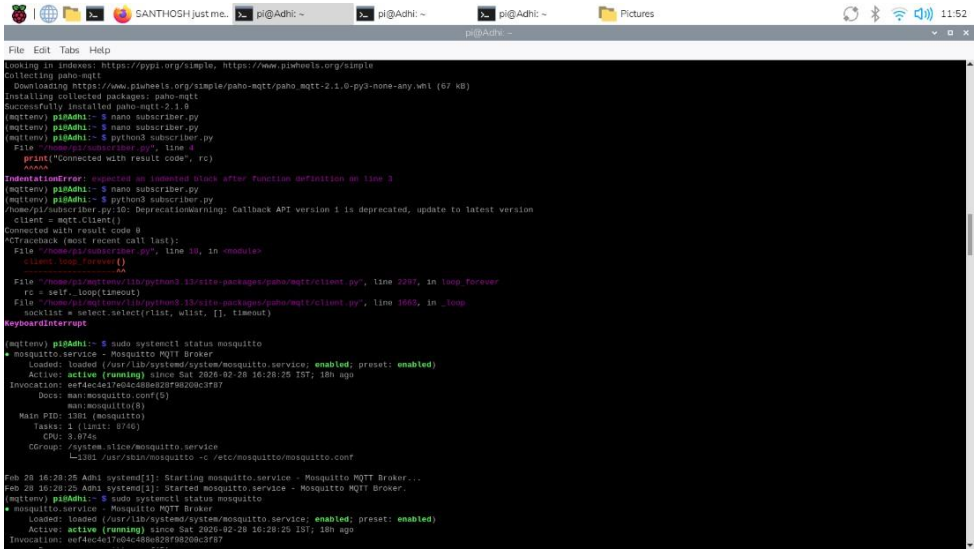
Step 3:

Checked and confirmed the Raspberry Pi IP address.



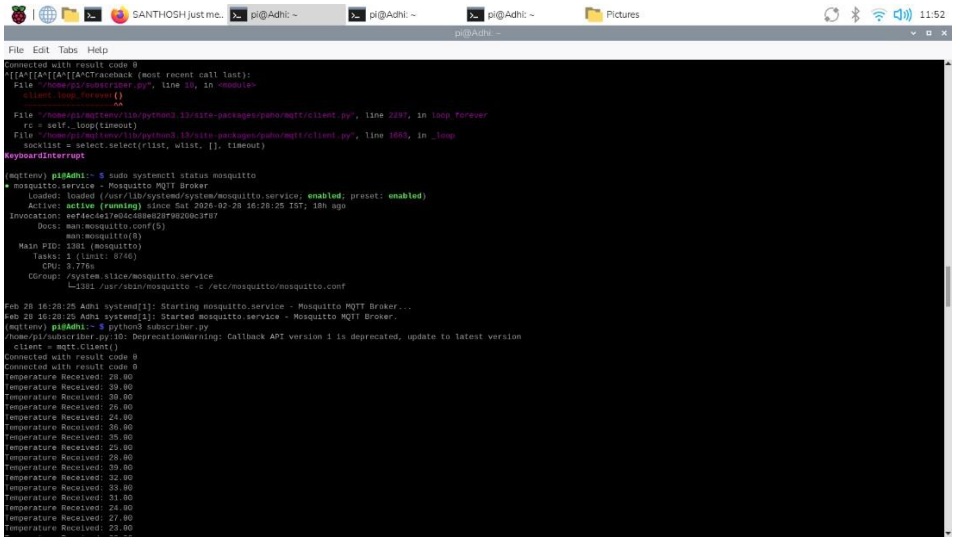
Step 4:

Installed MQTT library in Arduino IDE.



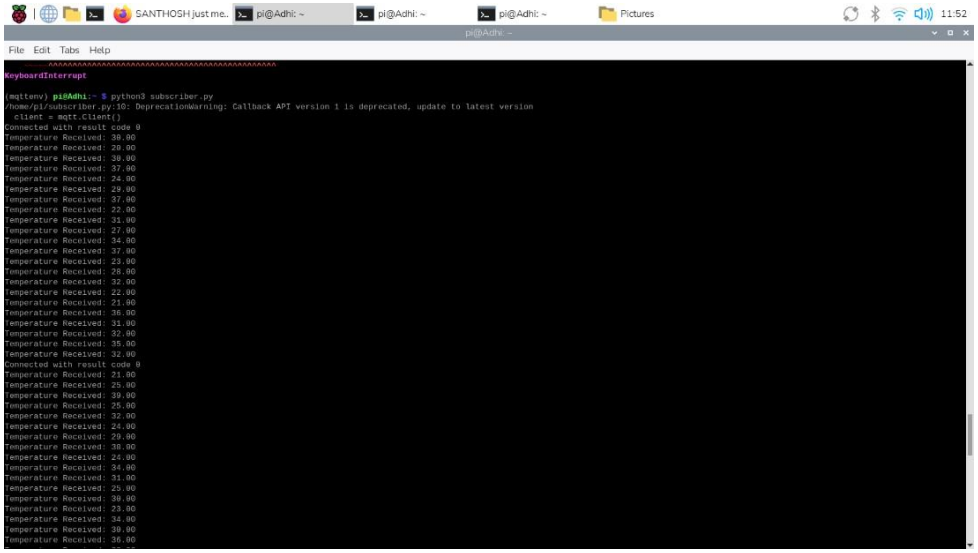
Step 5:

Opened and configured ESP32 publisher code in Arduino IDE.



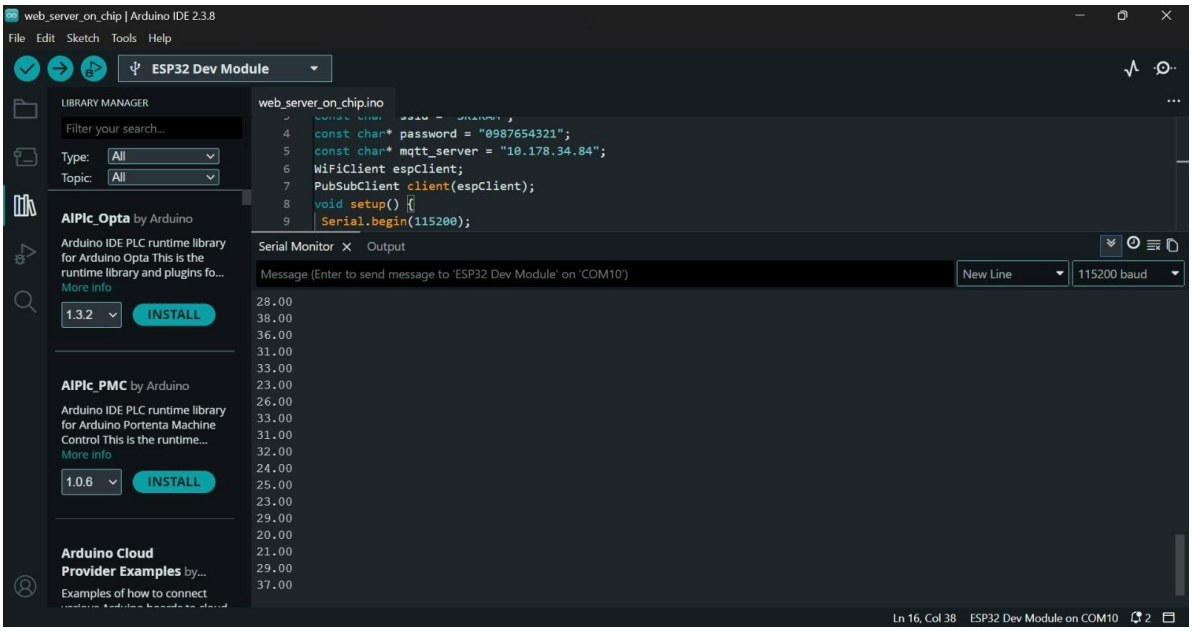
Step 6:

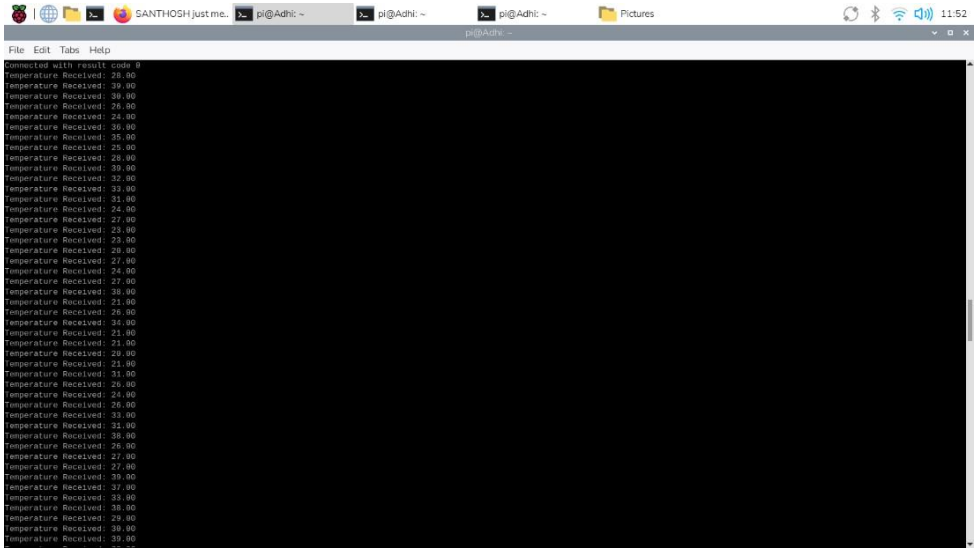
Uploaded the MQTT publisher program to ESP32 board.



Step 7:

Created Python subscriber program for receiving temperature data. Executed the Python MQTT subscriber script. ESP32 started publishing simulated temperature values. Raspberry Pi successfully received temperature data via MQTT. Verified continuous wireless communication between ESP32 and Raspberry Pi.




SANTHOSH just me..
pi@Adhi: ~
pi@Adhi: ~
pi@Adhi: ~
Pictures
11:52
File Edit Tabs Help
Connected with result code 0
Temperature Received: 28.00
Temperature Received: 39.00
Temperature Received: 30.00
Temperature Received: 26.00
Temperature Received: 24.00
Temperature Received: 36.00
Temperature Received: 35.00
Temperature Received: 25.00
Temperature Received: 28.00
Temperature Received: 39.00
Temperature Received: 32.00
Temperature Received: 33.00
Temperature Received: 31.00
Temperature Received: 24.00
Temperature Received: 27.00
Temperature Received: 23.00
Temperature Received: 23.00
Temperature Received: 28.00
Temperature Received: 27.00
Temperature Received: 24.00
Temperature Received: 27.00
Temperature Received: 38.00
Temperature Received: 21.00
Temperature Received: 26.00
Temperature Received: 36.00
Temperature Received: 21.00
Temperature Received: 21.00
Temperature Received: 28.00
Temperature Received: 21.00
Temperature Received: 31.00
Temperature Received: 26.00
Temperature Received: 24.00
Temperature Received: 26.00
Temperature Received: 33.00
Temperature Received: 31.00
Temperature Received: 36.00
Temperature Received: 26.00
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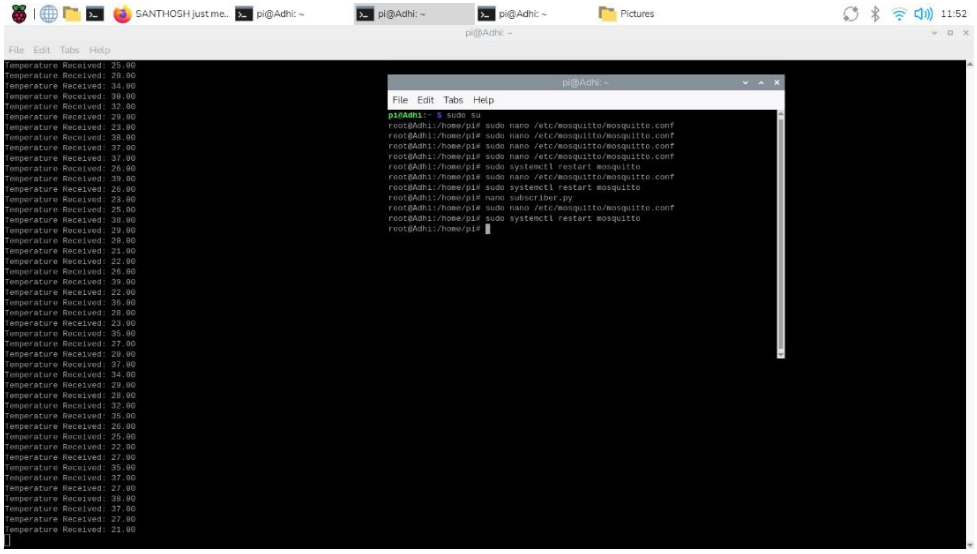
Temperature Received: 27.00
Temperature Received: 39.00
Temperature Received: 37.00
Temperature Received: 33.00
Temperature Received: 38.00
Temperature Received: 29.00
Temperature Received: 30.00
Temperature Received: 39.00


SANTHOSH just me..
pi@Adhi: ~
pi@Adhi: ~
pi@Adhi: ~
Pictures
11:52
File Edit Tabs Help
Temperature Received: 25.00
Temperature Received: 28.00
Temperature Received: 34.00
Temperature Received: 30.00
Temperature Received: 32.00
Temperature Received: 29.00
Temperature Received: 23.00
Temperature Received: 38.00
Temperature Received: 37.00
Temperature Received: 26.00
Temperature Received: 39.00
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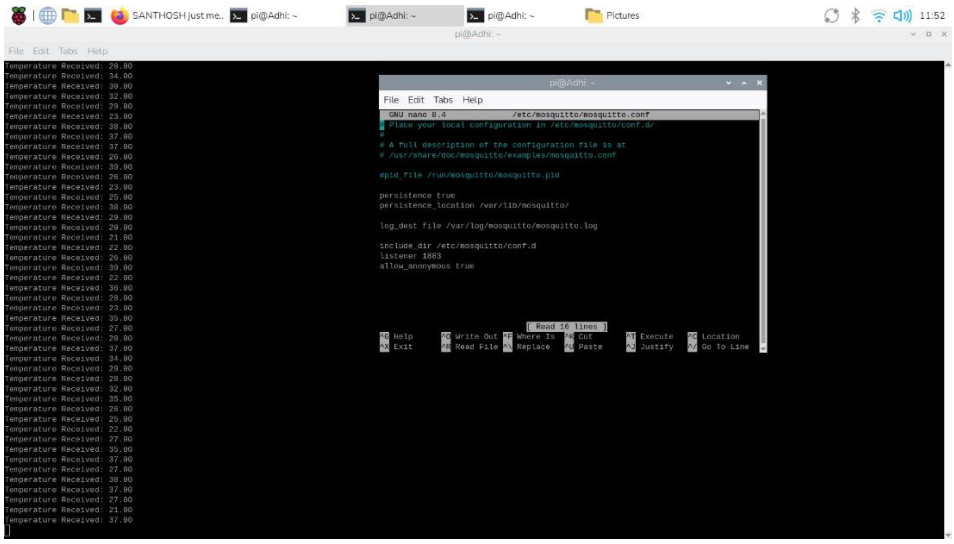
Temperature Received: 26.00
Temperature Received: 25.00
Temperature Received: 22.00
Temperature Received: 27.00
Temperature Received: 35.00
Temperature Received: 37.00
Temperature Received: 27.00
Temperature Received: 38.00
Temperature Received: 37.00
Temperature Received: 27.00
Temperature Received: 21.00
pi@Adhi: ~
File Edit Tabs Help
pi@Adhi:~ $ sudo su
root@Adhi:/home/pi# sudo nano /etc/mosquitto/mosquitto.conf
root@Adhi:/home/pi# sudo nano /etc/mosquitto/mosquitto.conf
root@Adhi:/home/pi# sudo nano /etc/mosquitto/mosquitto.conf
root@Adhi:/home/pi# sudo nano /etc/mosquitto/mosquitto.conf
root@Adhi:/home/pi# sudo systemctl restart mosquitto
root@Adhi:/home/pi# sudo nano /etc/mosquitto/mosquitto.conf
root@Adhi:/home/pi# sudo systemctl restart mosquitto
root@Adhi:/home/pi# nano subscriber.py
root@Adhi:/home/pi# sudo nano /etc/mosquitto/mosquitto.conf
root@Adhi:/home/pi# sudo systemctl restart mosquitto
root@Adhi:/home/pi#


SANTHOSH just me..
pi@Adhi: ~
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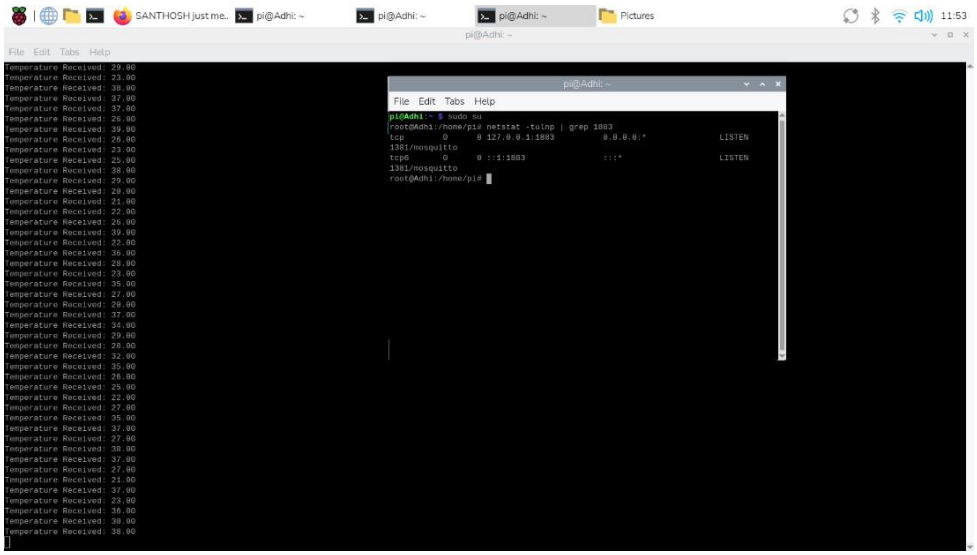
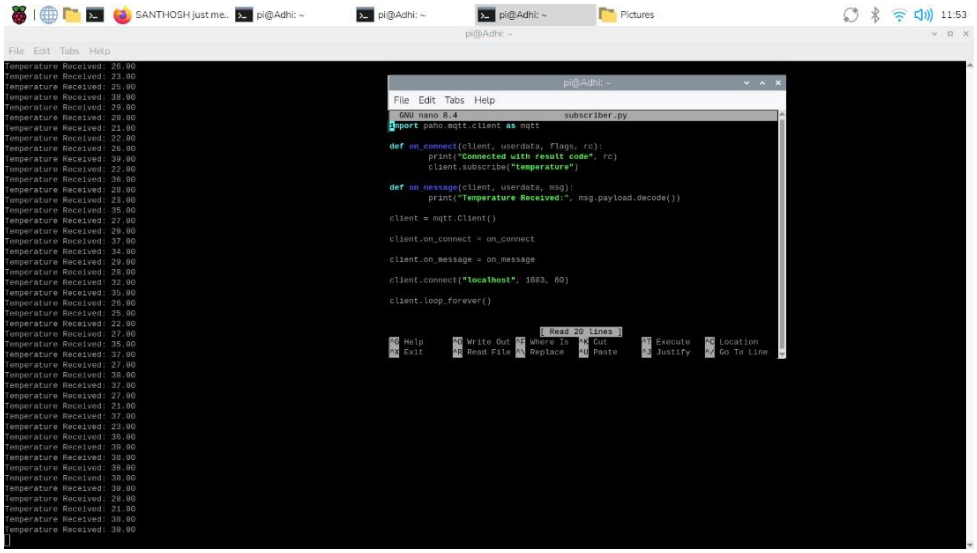
pi@Adhi: ~
Pictures
11:52
File Edit Tabs Help
Temperature Received: 28.00
Temperature Received: 34.00
Temperature Received: 30.00
Temperature Received: 32.00
Temperature Received: 29.00
Temperature Received: 23.00
Temperature Received: 38.00
Temperature Received: 37.00
Temperature Received: 37.00
Temperature Received: 26.00
Temperature Received: 39.00
Temperature Received: 26.00
Temperature Received: 23.00
Temperature Received: 25.00
Temperature Received: 38.00
Temperature Received: 29.00
Temperature Received: 28.00
Temperature Received: 21.00
Temperature Received: 22.00
Temperature Received: 26.00
Temperature Received: 39.00
Temperature Received: 22.00
Temperature Received: 36.00
Temperature Received: 28.00
Temperature Received: 23.00
Temperature Received: 35.00
Temperature Received: 27.00
Temperature Received: 28.00
Temperature Received: 37.00
Temperature Received: 34.00
Temperature Received: 29.00
Temperature Received: 28.00
Temperature Received: 32.00
Temperature Received: 35.00
Temperature Received: 26.00
Temperature Received: 25.00
Temperature Received: 22.00
Temperature Received: 27.00
Temperature Received: 35.00
Temperature Received: 37.00
Temperature Received: 27.00
Temperature Received: 38.00
Temperature Received: 37.00
Temperature Received: 27.00
Temperature Received: 21.00
pi@Adhi: ~
File Edit Tabs Help
GNU nano 8.4 /etc/mosquitto/mosquitto.conf
Place your local configuration in /etc/mosquitto/conf.d/
#
A full description of the configuration file is at
/usr/share/doc/mosquitto/examples/mosquitto.conf
#pid_file /run/mosquitto/mosquitto.pid
persistence true
persistence_location /var/lib/mosquitto/
log_dest file /var/log/mosquitto/mosquitto.log
include_dir /etc/mosquitto/conf.d
listener 1883
allow_anonymous true
[Read 16 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

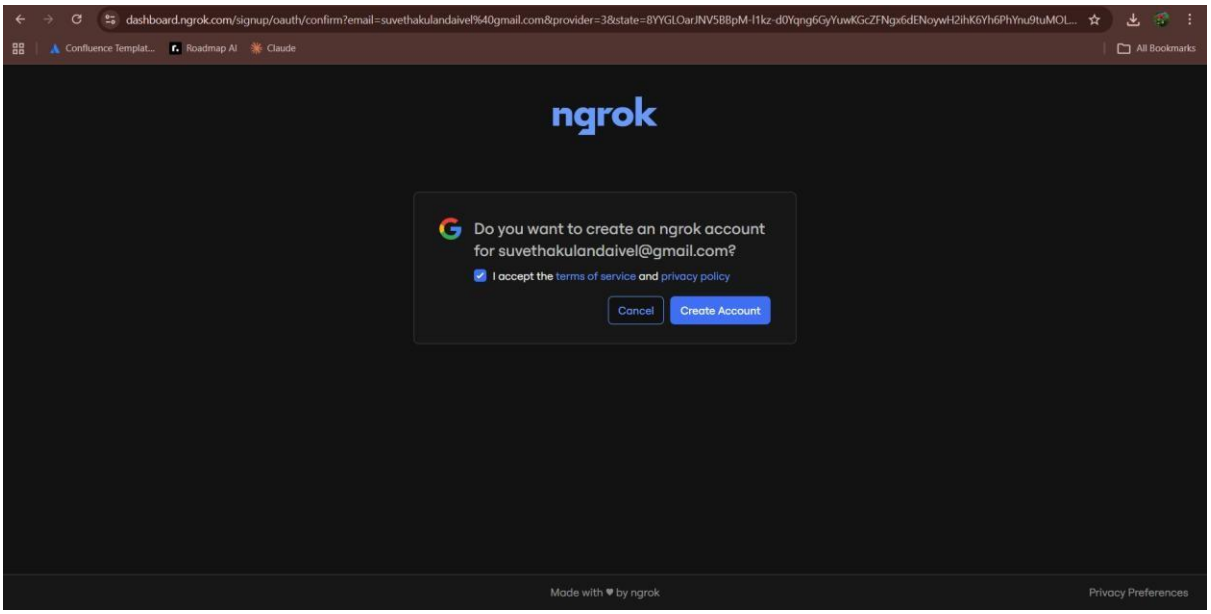


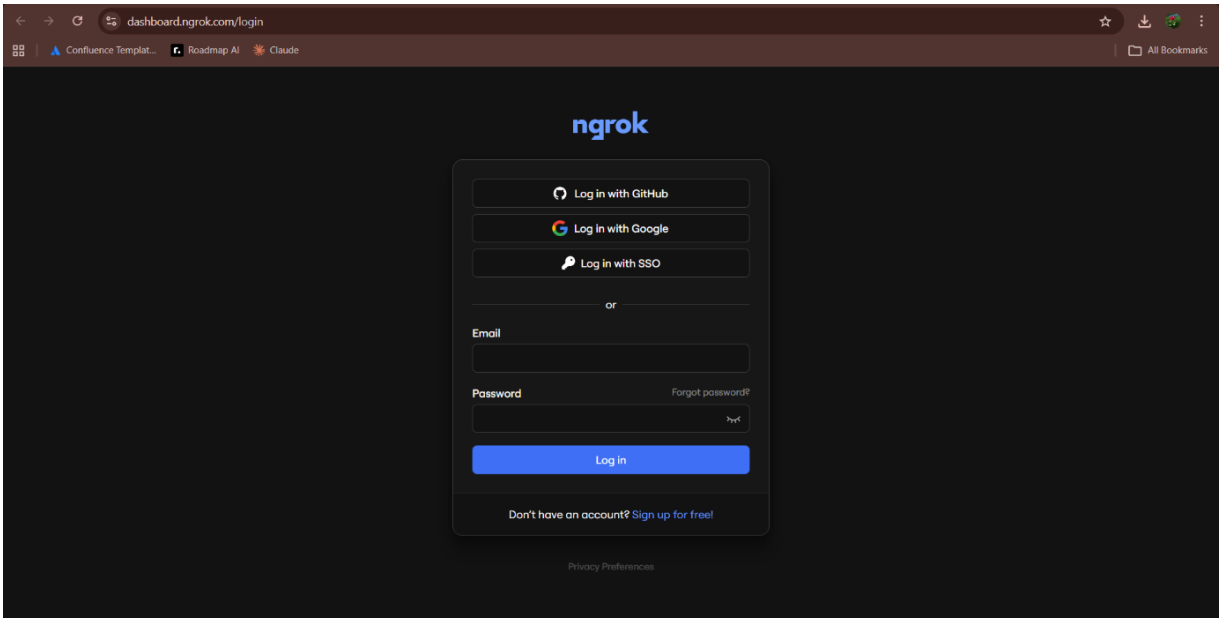


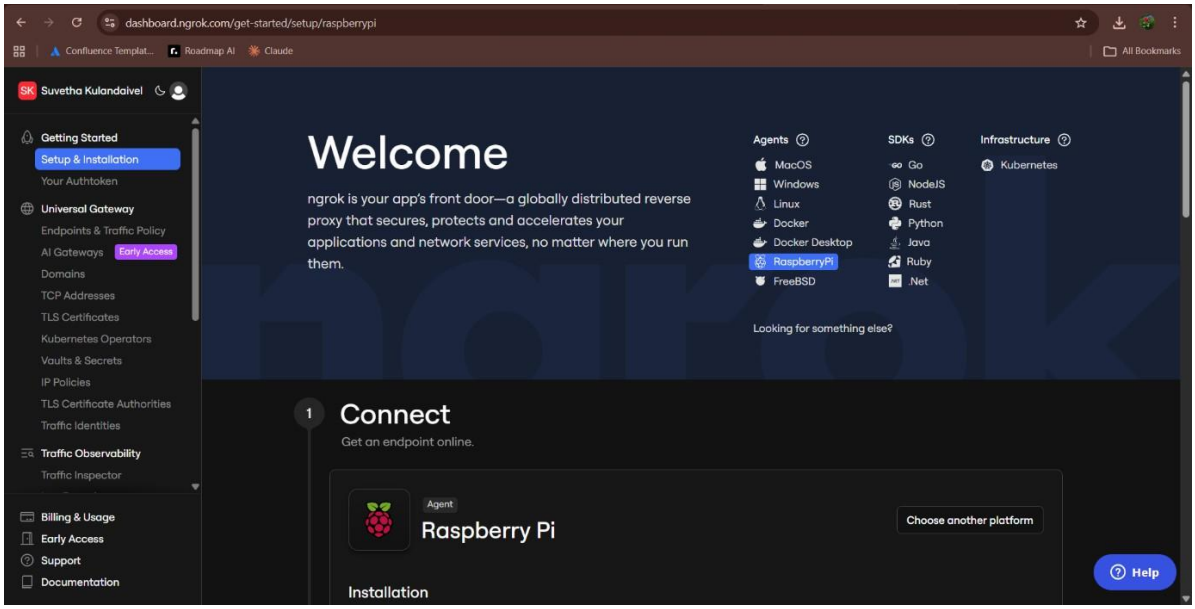
Project 2

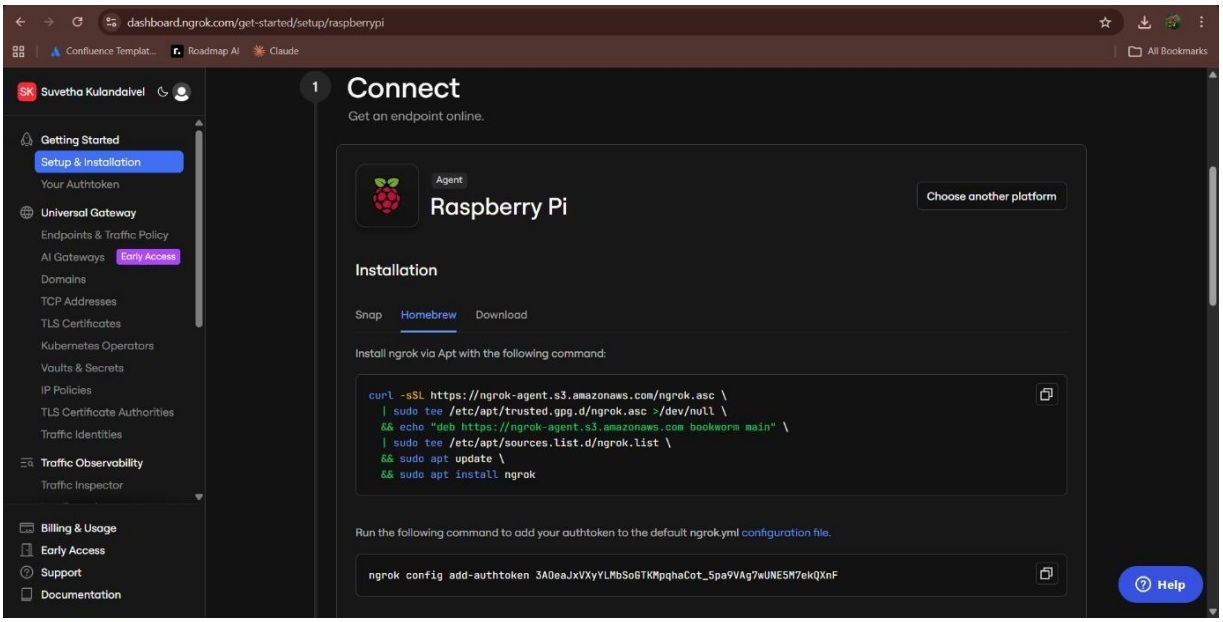
Prodecure:

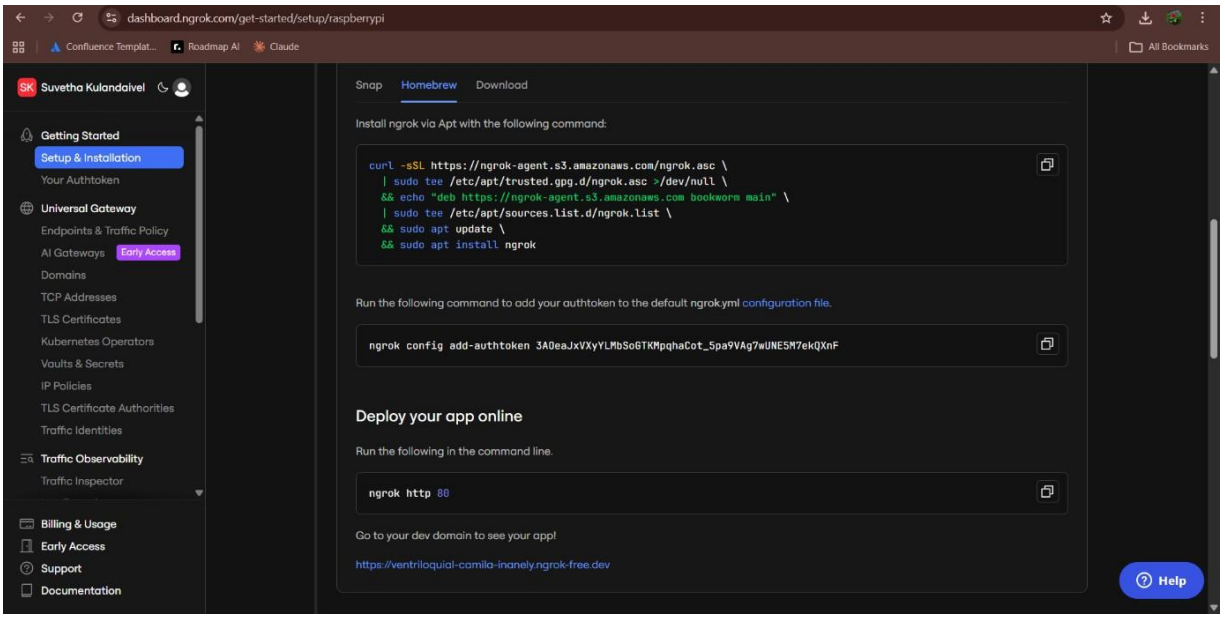
- ✓ Login to ngrok
- ✓ Install ngrok on Raspberry Pi (via apt)
- ✓ Add your authtoken (ngrok config add-authtoken token)
- ✓ Start your local server
- ✓ Run ngrok http 80
- ✓ Use the generated public URL.


dashboard.ngrok.com/signup/oauth/confirm?email=suvethakulandaivel%40gmail.com&provider=3&state=8YYGLOarJNV5BBpM-I1kz-d0Yqng6GyYuwKGcZFNgx6dENoywH2ihK6Yh6PhYnu9tuMOL...
Confluence Templat...
Roadmap AI
Claude
All Bookmarks
ngrok
Do you want to create an ngrok account for suvethakulandaivel@gmail.com?
I accept the terms of service and privacy policy
Cancel
Create Account
Made with ♥ by ngrok
Privacy Preferences


dashboard.ngrok.com/login
Confluence Templat...
Roadmap AI
Claude
All Bookmarks
ngrok
Log in with GitHub
Log in with Google
Log in with SSO
or
Email
Password
Forgot password?
Log in
Don't have an account? Sign up for free!
Privacy Preferences


dashboard.ngrok.com/get-started/setup/raspberrypi
Confluence Templat...
Roadmap AI
Claude
All Bookmarks
SK Suvetha Kulandaivel
Getting Started
Setup & Installation
Your Authtoken
Universal Gateway
Endpoints & Traffic Policy
AI Gateways
Early Access
Domains
TCP Addresses
TLS Certificates
Kubernetes Operators
Vaults & Secrets
IP Policies
TLS Certificate Authorities
Traffic Identities
Traffic Observability
Traffic Inspector
Billing & Usage
Early Access
Support
Documentation
Welcome
ngrok is your app's front door—a globally distributed reverse proxy that secures, protects and accelerates your applications and network services, no matter where you run them.
Agents
MacOS
Windows
Linux
Docker
Docker Desktop
RaspberryPi
FreeBSD
SDKs
Go
NodeJS
Rust
Python
Java
Ruby
.Net
Infrastructure
Kubernetes
Looking for something else?
1
Connect
Get an endpoint online.
Agent
Raspberry Pi
Choose another platform
Installation
Help


dashboard.ngrok.com/get-started/setup/raspberrypi
Confluence Templat...
Roadmap AI
Claude
All Bookmarks
Suvetha Kulandaivel
Getting Started
Setup & Installation
Your Authtoken
Universal Gateway
Endpoints & Traffic Policy
AI Gateways
Early Access
Domains
TCP Addresses
TLS Certificates
Kubernetes Operators
Vaults & Secrets
IP Policies
TLS Certificate Authorities
Traffic Identities
Traffic Observability
Traffic Inspector
Billing & Usage
Early Access
Support
Documentation
1
Connect
Get an endpoint online.
Agent
Raspberry Pi
Choose another platform
Installation
Snap
Homebrew
Download
Install ngrok via Apt with the following command:
curl -sSL https://ngrok-agent.s3.amazonaws.com/ngrok.asc \
| sudo tee /etc/apt/trusted.gpg.d/ngrok.asc >/dev/null \
&& echo "deb https://ngrok-agent.s3.amazonaws.com bookworm main" \
| sudo tee /etc/apt/sources.list.d/ngrok.list \
&& sudo apt update \
&& sudo apt install ngrok
Run the following command to add your authtoken to the default ngrok.yml configuration file.
ngrok config add-authtoken 3AOeaJxVXyYLMbSoGTKMpqhaCot_5pa9VAg7wUNE5M7ekQXnF
Help


dashboard.ngrok.com/get-started/setup/raspberrypi
Confluence Templat...
Roadmap AI
Claude
All Bookmarks
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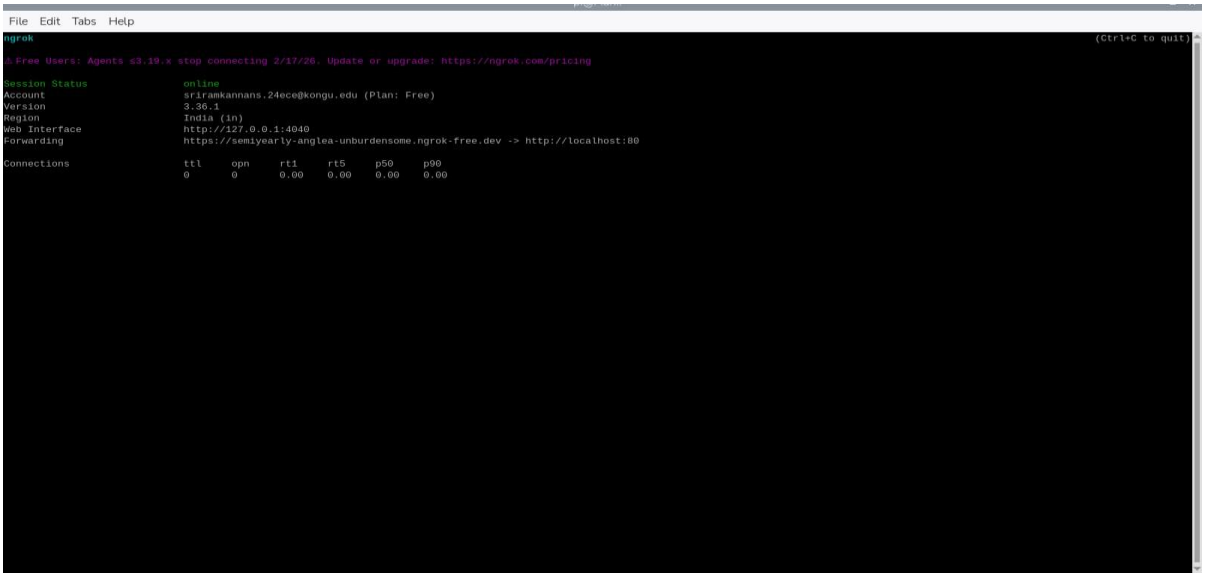
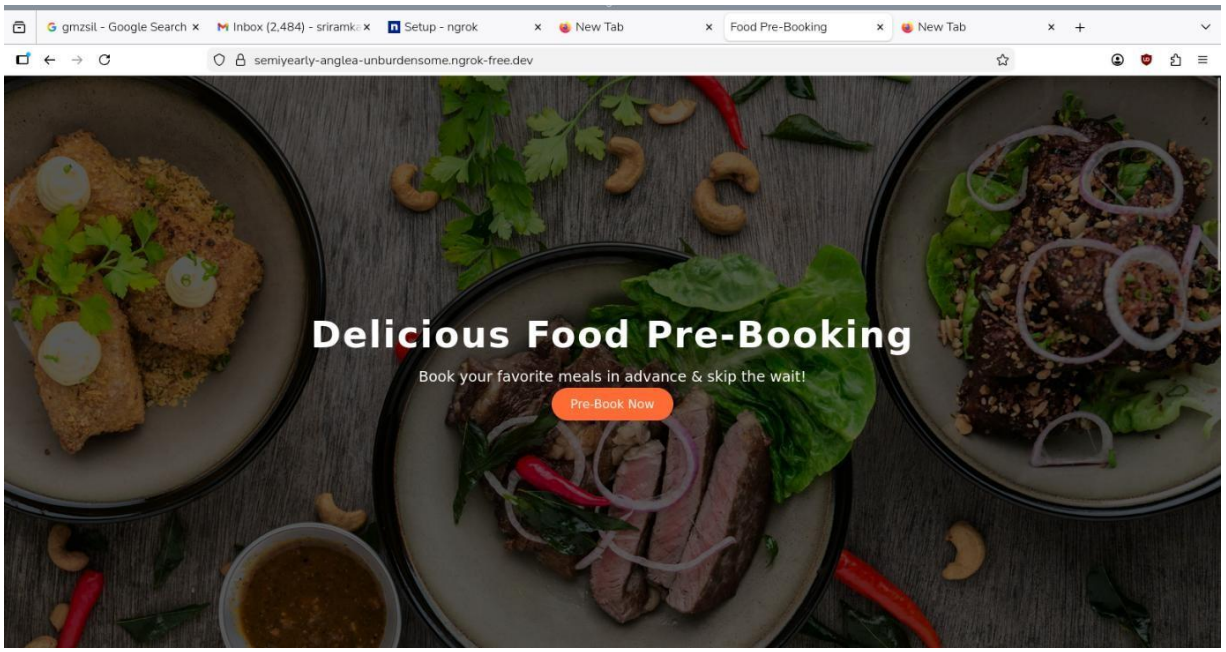
Getting Started
Setup & Installation
Your Authtoken
Universal Gateway
Endpoints & Traffic Policy
AI Gateways
Early Access
Domains
TCP Addresses
TLS Certificates
Kubernetes Operators
Vaults & Secrets
IP Policies
TLS Certificate Authorities
Traffic Identities
Traffic Observability
Traffic Inspector
Billing & Usage
Early Access
Support
Documentation
Snap
Homebrew
Download
Install ngrok via Apt with the following command:
curl -sSL https://ngrok-agent.s3.amazonaws.com/ngrok.asc \
| sudo tee /etc/apt/trusted.gpg.d/ngrok.asc >/dev/null \
&& echo "deb https://ngrok-agent.s3.amazonaws.com bookworm main" \
| sudo tee /etc/apt/sources.list.d/ngrok.list \
&& sudo apt update \
&& sudo apt install ngrok
Run the following command to add your authtoken to the default ngrok.yml configuration file.
ngrok config add-authtoken 3AOeaJxVXyYLMbSoGTKMpqhaCot_5pa9VAg7wUNE5M7ekQXnF
Deploy your app online
Run the following in the command line.
ngrok http 80
Go to your dev domain to see your app!
https://ventriloquial-camila-inanely.ngrok-free.dev
Help

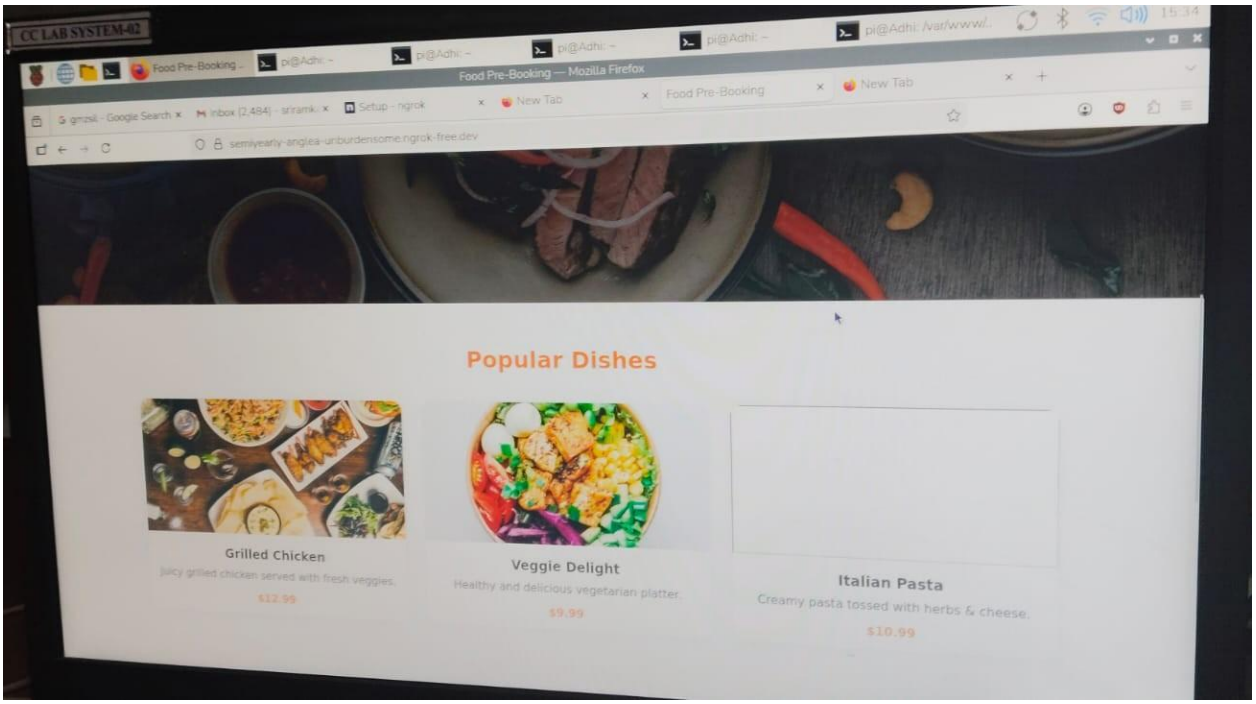

File Edit Tabs Help
ngrok
(Ctrl+C to quit)
Free Users: Agents s3.19.x stop connecting 2/17/26. Update or upgrade: https://ngrok.com/pricing
Session Status online
Account sriramkannans.24ece@kongu.edu (Plan: Free)
Version 3.36.1
Region India (in)
Web Interface http://127.0.0.1:4040
Forwarding https://semiyearly-anglea-unburdensome.ngrok-free.dev -> http://localhost:80
Connections ttl opn rt1 rt5 p50 p90
0 0 0.00 0.00 0.00 0.00

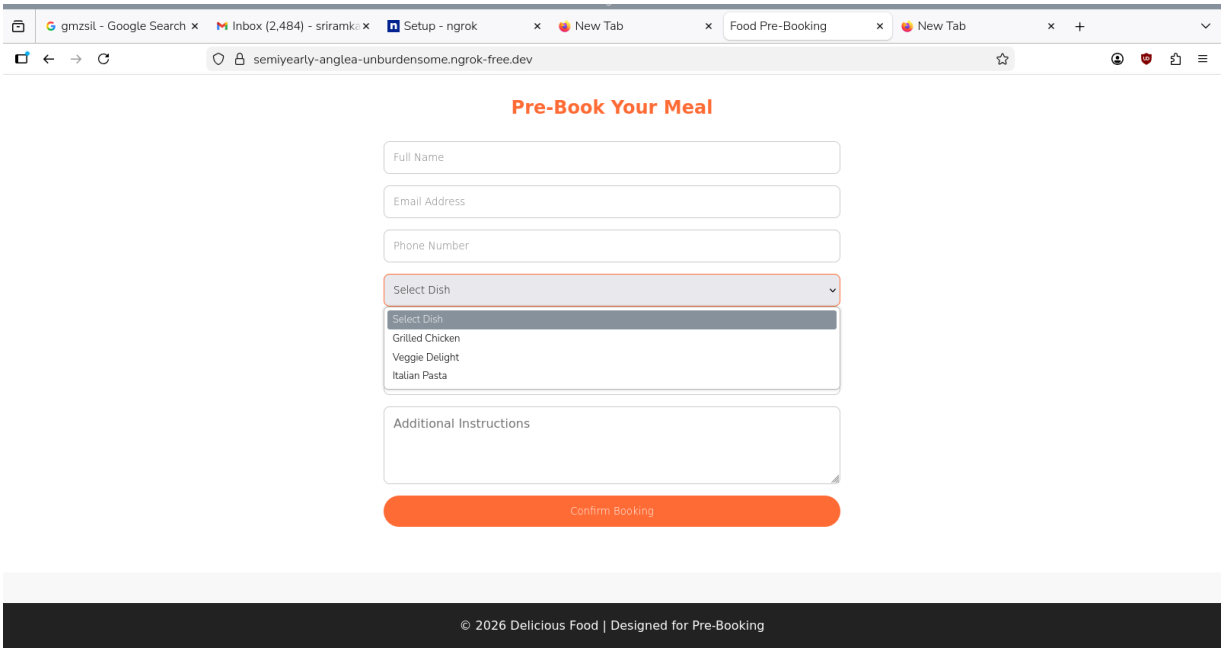
Web Link:

https://semiyearly-anglea-unburdensome.ngrok-free.dev/

Web Output:







Result:

The ESP32 successfully published temperature data and Raspberry Pi received it using MQTT protocol. This demonstrates wireless IoT communication.